Photosynthesis CER Cheat Sheet

The science, the lab, the CER method and the traps. Sources: [P1] POGIL worksheet · [P2] simulation lab · [P3] Mauna Loa notes

A The two equations: energy IN vs energy OUT

[P1 Model 1 · Q1–2 · Q14]

PHOTOSYNTHESIS in the **chloroplast** · chlorophyll captures the light

$6CO_2 + 6H_2O +$ **light energy** (IN) $\rightarrow C_6H_{12}O_6 + 6O_2$

light energy is changed into chemical energy stored in glucose

CELLULAR RESPIRATION in the **mitochondrion** · runs 24/7, light or dark

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O +$ **energy (ATP)** (OUT)

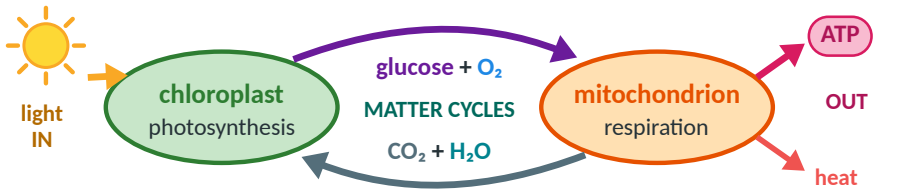
releases the energy stored in glucose as ATP, the cell's usable energy

#1 WATCH-OUT: don't swap the energy labels!

- **Light** energy goes **IN** to photosynthesis.
- **ATP** energy comes **OUT** of respiration.
- Same molecules, reversed: only the energy differs [P1 Q9].

B One cycle: matter cycles, energy flows

[P1 Model 1 · Q3–Q10]



ENERGY FLOWS one way: **light** → glucose → **ATP** → **heat** (not recycled) [P1 Q4]

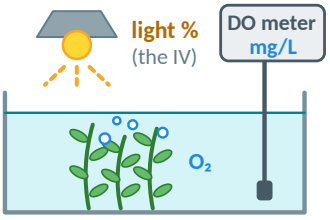
WHO DOES WHAT

	photo-synthesis	respi-ration
Plants chloroplasts + mitochondria	✓	✓
Animals mitochondria, **no** chloroplasts	✗	✓
Decomposers fungi, bacteria, worms	✗	✓

Respiration ≠ breathing: it is a reaction in cells.

C The lab in one picture: light intensity and O_2

[P2 Parts II–III]



DO = dissolved O_2 in the water

EXPERIMENT DESIGN

IV (changed): light intensity (%)
DV (measured): O_2 production
Same: 0 fish, 1 plant, 25 °C, white light, 60-min run
Control: Part II dark run (respiration only, 3 trials)
Limitation: only 1 trial per light level

TWO NUMBERS: DON'T MIX THEM UP

DO difference = DO end − DO start
= **net** change = O_2 made − O_2 used

O_2 production = DO difference − respiration
= **total** O_2 made by photosynthesis (**gross**)

SIGN TRAP: respiration = **−0.1939**, so DO diff − (−0.1939) = **DO diff + 0.1939**

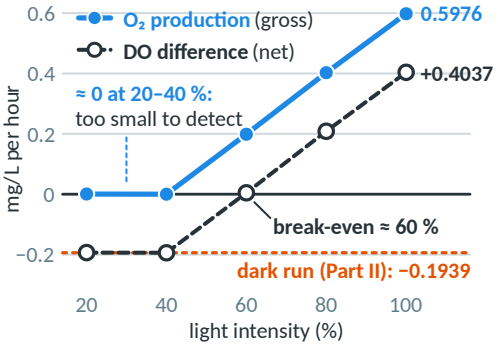
At 100 % light (mg/L per hour):

+0.4037 kept (DO difference) | 0.1939 used

= **0.5976 made = O_2 production** (gross)

per minute: 0.5976 ÷ 60 = **0.0100 mg/L per minute**

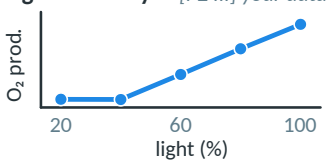
YOUR PART III DATA — 1 trial per level



D What controls the rate: limiting factors

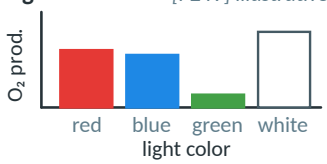
Parts IV–V graphs are illustrative: get the real numbers from classmates

Light intensity *[P2 III] your data*



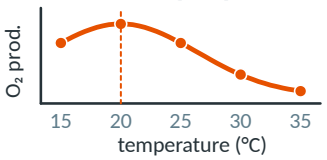
From 40 % up, each +20 % light ≈ **+0.20 mg/L per hour** more O_2. No leveling off: light was limiting.

Light color *[P2 IV] illustrative*



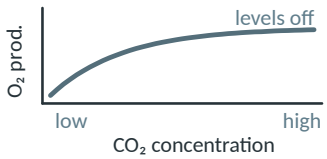
White was best (class results). Chlorophyll absorbs **red + blue/violet**, reflects **green**.

Temperature *[P2 V] illustrative*



Best tested ≈ **20 °C** (optimum 15–25 °C). Warmer: enzymes less efficient; respiration rises.

CO_2 level *inference · illustrative*



CO_2 is a reactant: more CO_2 → faster, until another factor limits.

E CER = Claim + Evidence + Reasoning: 5 steps, scored /6

[P2 Part VII] label AND color: [C] blue · [E] green · [R] orange

THE 5-STEP CER METHOD

1. **Unpack the question:** circle the variables (what changed? what was measured?) and the command word.
2. **Read the data:** find the trend, the highest/lowest, and 2–3 numbers **with units**; note anything odd.
3. **Claim it [C]:** one sentence that answers the question in its own words.
4. **Prove it (Evidence) [E]:** 2–3 specific data points with units, ideally a comparison ("from __ to __").
5. **Explain it (Reasoning) [R]:** "This happens because… Since… therefore…" Name reactants, products, organelle, energy, and (if relevant) respiration happening at the same time.

SCORING /6

(typical; your teacher's may differ a bit)

[C] 1 pt | **[E] Evidence: 2 pts** | **[R] Reasoning: 3 pts**

[C] 1 = accurate, complete, answers the question (no "I think")
[E] 2 = ≥ 2 accurate data points with units (1 = vague or only one)
[R] 3 = right science principle + explicit link to the claim + vocabulary (2 = weak link · 1 = repeats the evidence · 0 = wrong science)

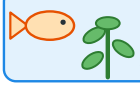
CLAIM TEMPLATES

Trend: As [IV] increases, [DV] increases/decreases.
Best condition: [Condition] gives the most [DV].
Comparison: [A] produced more [DV] than [B].
Mechanism: [Pattern] happens because [process].
Model-based: [Action] will increase/decrease [quantity].

F Reasoning toolkit: science to cite in your [R]

[P1] [P2] [P3]: use the packets' own words

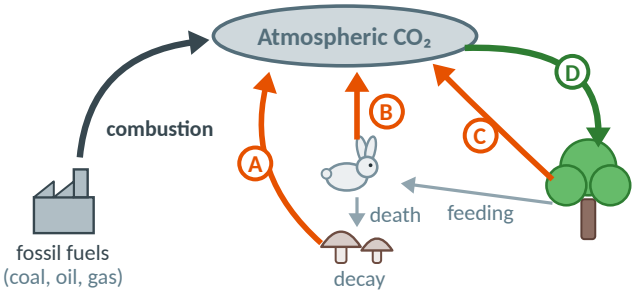
1. Photosynthesis stores **light energy** as chemical energy in **glucose**.
2. **O_2 is a product:** faster photosynthesis → more O_2 released.
3. Respiration breaks down glucose + O_2 to release energy as **ATP**.
4. **Plants do both.** Respiration runs 24/7, so in the dark DO falls.
5. DO difference = O_2 made − O_2 used (> 0 only if photosynthesis wins).
6. **Light is the energy source:** more light → faster photosynthesis.
7. Chlorophyll absorbs **red + blue/violet** light best and **reflects green**.
8. Enzymes work best at an **optimum temperature**.
9. The factor in shortest supply (the **limiting factor**) sets the rate.
10. Air CO_2: respiration + burning **add** it; photosynthesis **removes** it.



PART VII FISH LOGIC [P2 Part VII]: fish only respire, so the tank stays healthy only while **DO difference ≥ 0**. One plant at 100 % light left a **+0.4037 mg/L per hour** surplus → the fish O_2 budget is **≈ 0.40 mg/L per hour** for all the fish together. To find the number: add fish one at a time, rerun, and keep the last number with DO difference ≥ 0.

G The carbon cycle

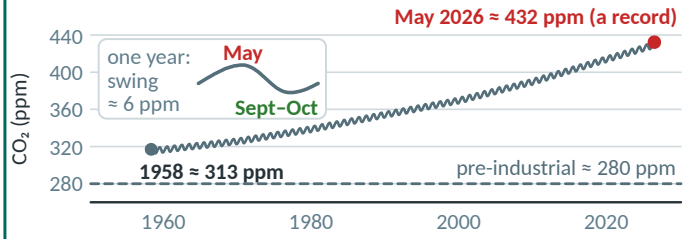
[P1 Model 2]



D = photosynthesis: takes CO_2 **out** of the air
A, B, C = respiration (decomposers, animals, plants): puts CO_2 **in**
Humans add CO_2 fast: burning fossil fuels (carbon stored for millions of years) + deforestation. No humans: in ≈ out [P1 Q18–20].

H The Keeling Curve (Mauna Loa)

[P3] · illustrative



Zig-zag = Northern-Hemisphere plants "breathing". CO_2 is a **running total, like water in a bathtub**: fall + winter, respiration + decay add more than photosynthesis removes → **peak in May**; late May through summer, leaves draw it down → **low in Sept–Oct**.

Long rise = burning fossil fuels (since the 1800s; faster after the 1950s): more is added each year than removed (≈ +2–2.5 ppm per year).

I Top 12 watch-outs (and the fix)

the traps that cost points

1 **Swapping the energy labels**	→ **Light** energy goes **IN** to photosynthesis; **ATP** comes **OUT** of respiration [P1 Q14].
2 **Forgetting that plants also respire**	→ Plants have chloroplasts **AND** mitochondria, so they respire 24/7, light or dark.
3 **Saying animals lack mitochondria**	→ Animals lack **chloroplasts**. They **do** have mitochondria [P1 Q10].
4 **Mixing up DO difference and O_2 production**	→ DO difference = **net**; O_2 production = **gross** = DO difference **+ 0.1939** (minus a negative = plus).
5 **Evidence without numbers or units**	→ Quote 2–3 numbers with units, ideally a comparison: "from 0.0011 to 0.5976 mg/L per hour".
6 **Reasoning that just repeats the evidence**	→ Say **why** with science: the process, reactants/products, organelle, energy.
7 **Claiming beyond the data**	→ Light above 100 % was never tested: don't claim "200 % light would be even better".
8 **Keeping the mistaken notes [P3]**	→ Respiration is **not** negligible; photosynthesis is **not** "delayed": CO_2 is a running total.
9 **"Respiration = breathing"**	→ Breathing moves gases in and out of the lungs; cellular respiration is a reaction in mitochondria.
10 **"Enzymes denature just above 20 °C"**	→ Above the optimum they just work less well; real denaturing needs ≈ 40 °C or more.
11 **Graph doesn't match the table**	→ IV on x, DV on y, even scale, units, mark the points, a title; don't draw beyond the data.
12 **Forgetting units and the time basis**	→ Write "mg/L per hour" (60-min run); per minute = ÷ 60 (0.5976 → 0.0100 mg/L per minute).